New Section

Problem Statement: Walmart wants to analyze customer purchase behavior during Black Friday sales to check if spending habits differ based on gender, marital status, and age group. The aim is to understand whether women spend more than men and how other demographic factors influence total spending.

Goal:

The goal of this analysis is to explore Walmart's sales data to find patterns in customer spending across gender, marital status, and age groups. Using statistical tools like Confidence Intervals and the Central Limit Theorem, we will estimate and compare average expenditure for each group and provide clear business recommendations for targeted marketing and better decision-making.

Q1:Import the dataset and do usual data analysis steps like checking the structure & characteristics of the dataset.

In [ ]:
```
import pandas as pd

# Load dataset, skipping problematic lines
df = pd.read_csv('walmart_data.csv', on_bad_lines='skip')
df.head()                              # see first 5 rows
```

Out[ ]:

	User_ID	Product_ID	Gender	Age	Occupation	City_Category	Stay_In_Current_City_Years	Marital_Status	Product_Category	Purchase
0	1000001	P00069042	F	0-17	10	A	2	0	3	8370
1	1000001	P00248942	F	0-17	10	A	2	0	1	15200
2	1000001	P00087842	F	0-17	10	A	2	0	12	1422
3	1000001	P00085442	F	0-17	10	A	2	0	12	1057
4	1000002	P00285442	M	55+	16	C	4+	0	8	7969

In [ ]:
```
df.shape
```

Out[ ]:
```
(550068, 10)
```

Convert Categorical Columns to 'category'

In [ ]:
```
for col in ['Gender','Age','City_Category','Stay_In_Current_City_Years','Product_Category']:
    df[col] = df[col].astype('category')
```

In [ ]:
```
df.describe()
```

Out[ ]:

	User_ID	Occupation	Marital_Status	Purchase
count	5.500680e+05	550068.000000	550068.000000	550068.000000
mean	1.003029e+06	8.076707	0.409653	9263.968713
std	1.727592e+03	6.522660	0.491770	5023.065394
min	1.000001e+06	0.000000	0.000000	12.000000
25%	1.001516e+06	2.000000	0.000000	5823.000000
50%	1.003077e+06	7.000000	0.000000	8047.000000
75%	1.004478e+06	14.000000	1.000000	12054.000000
max	1.006040e+06	20.000000	1.000000	23961.000000

Non-Graphical Analysis (Value Counts & Unique Attributes)

In [ ]:
```
for col in df.columns:
    print(f"{col}: {df[col].nunique()} unique values")
```

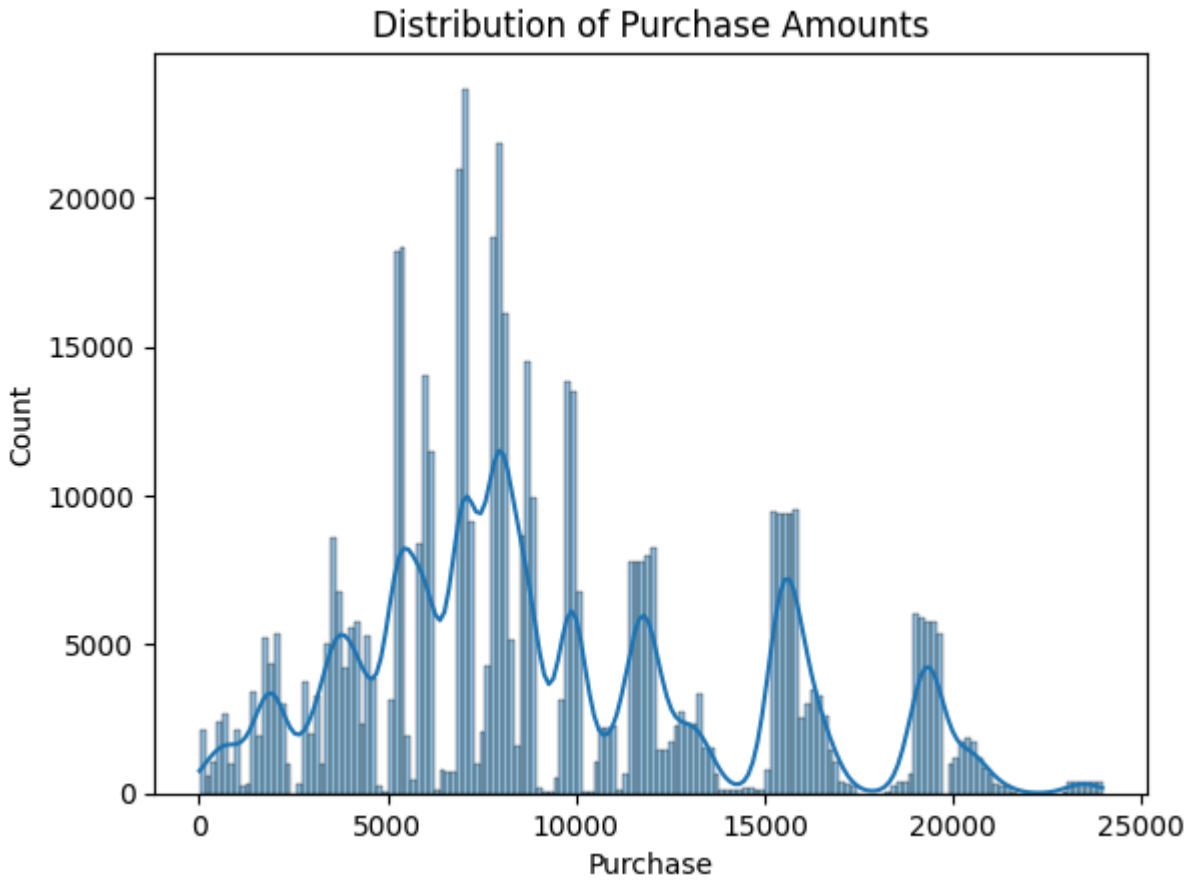
```
User_ID: 5891 unique values
Product_ID: 3631 unique values
Gender: 2 unique values
Age: 7 unique values
Occupation: 21 unique values
City_Category: 3 unique values
Stay_In_Current_City_Years: 5 unique values
Marital_Status: 2 unique values
Product_Category: 20 unique values
Purchase: 18105 unique values
Age_Bin: 5 unique values
```

1.4 Visual Analysis

Univariate Analysis

For continuous variables (Purchase):

In [ ]:
```
sns.histplot(df['Purchase'], kde=True)
plt.title("Distribution of Purchase Amounts")
plt.show()
```



Univariate Analysis — Purchase Amount

Observation:

Purchase amounts are right-skewed.

Most customers spend between ₹5,000 – ₹12,000.

A few customers spend as high as ₹24,000.

Interpretation:

High-value purchases are rare.

Data shows multiple spending peaks (multimodal).

A small segment of customers makes very large purchases.

Business Insight:

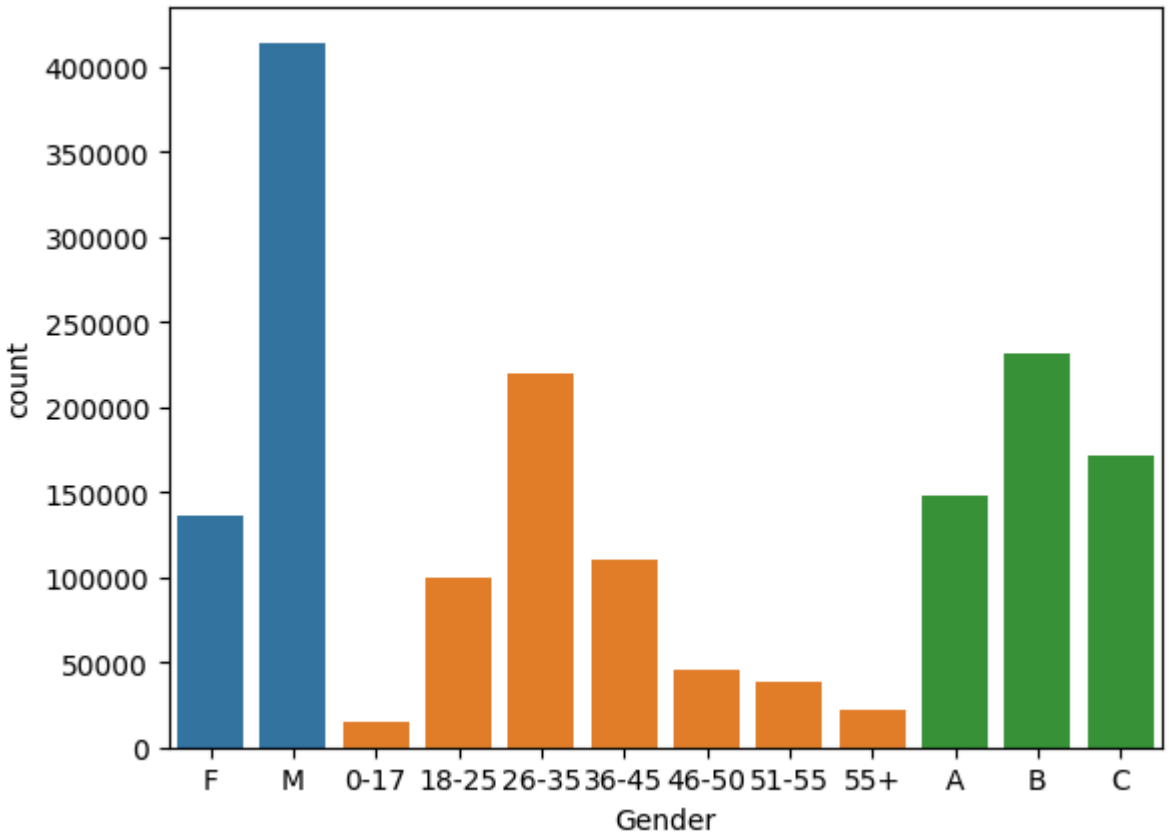
Walmart's customer base is mostly medium spenders.

Targeting premium customers (high spenders) can increase total revenue.

Introduce exclusive offers or loyalty programs for big spenders.

For categorical variables (Gender, Age, City_Category):

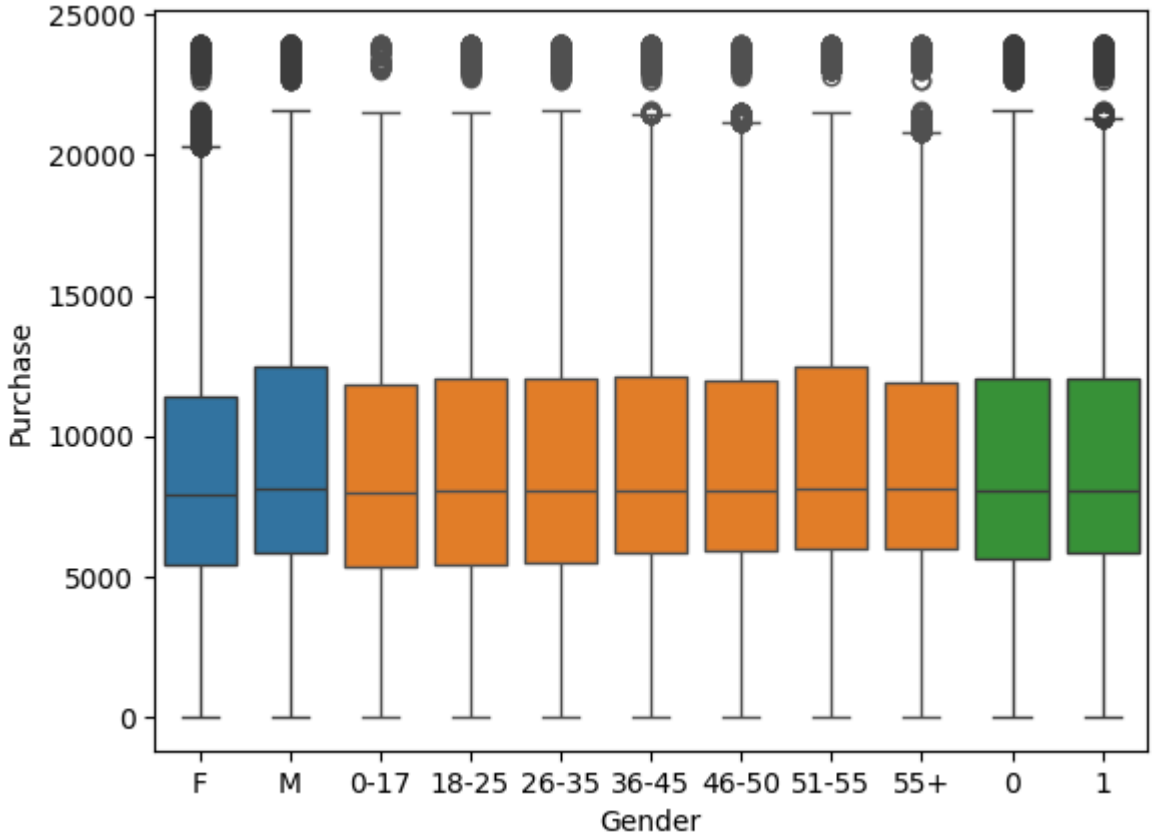
In [ ]:
```
sns.countplot(x='Gender', data=df)
sns.countplot(x='Age', data=df)
sns.countplot(x='City_Category', data=df)
plt.show()
```



Bivariate Analysis

Purchase vs Gender / Age / Marital_Status:

In [ ]:
```
sns.boxplot(x='Gender', y='Purchase', data=df)
sns.boxplot(x='Age', y='Purchase', data=df)
sns.boxplot(x='Marital_Status', y='Purchase', data=df)
plt.show()
```



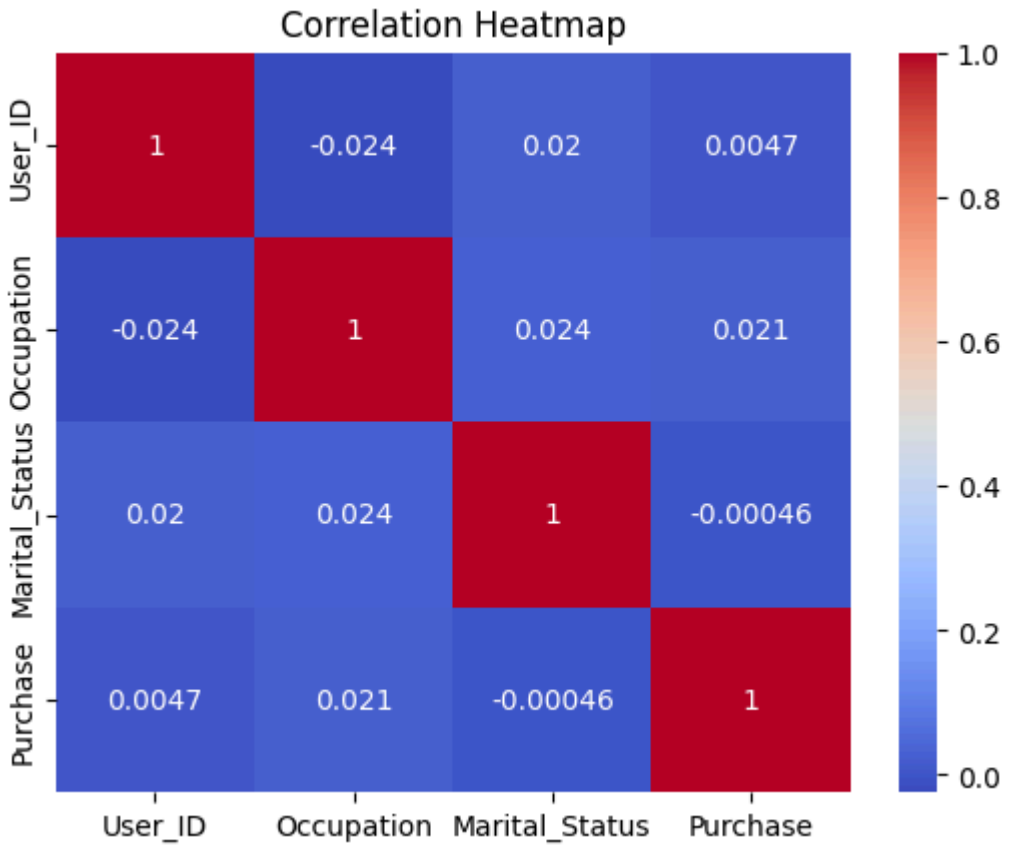
Bivariate Analysis — Interpretation

Male customers and the 26–35 age group show higher median purchase amounts, while married and unmarried customers spend almost the same on average.

This indicates that gender and age influence spending behavior, but marital status has little or no effect on purchase amounts.

Correlation (for numeric variables)

In [ ]:
```
num = df.select_dtypes(include=['number'])
sns.heatmap(num.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



Correlation Analysis — Interpretation

Correlation values between Purchase and other numeric variables are very low (near 0).

This means Occupation and Marital_Status have almost no linear relationship with spending.

Customer demographics (like Gender, Age, City) likely influence purchase behavior more than numeric factors.

Walmart should focus on demographic-based marketing, not occupation or marital status, for better targeting.

Q2:Detect Null values & Outliers (using boxplot, "describe" method by checking the difference between mean and median, isnull etc.)

In [ ]:
```
# Check missing values
df.isnull().sum()
```

Out[ ]:

	0
User_ID	0
Product_ID	0
Gender	0
Age	0
Occupation	0
City_Category	0
Stay_In_Current_City_Years	0
Marital_Status	0
Product_Category	0
Purchase	0

dtype: int64

In [ ]:
```
# Quick numeric summary to compare mean vs median
df['Purchase'].describe()
```

Out[ ]:

	Purchase
count	550068.000000
mean	9263.968713
std	5023.065394
min	12.000000
25%	5823.000000
50%	8047.000000
75%	12054.000000
max	23961.000000

dtype: float64

The average purchase amount is around ₹9,241, and most customers spend between ₹5,800 and ₹12,000. Since the mean is higher than the median, a few customers spent much more — showing right-skewed data (some high spenders).

In [ ]:
```
# Value counts and unique values for each column
for col in df.columns:
    print(f"\nColumn: {col}")
    print("Unique values:", df[col].nunique())
    if df[col].dtype == 'category' or df[col].dtype == 'object':
        print(df[col].value_counts().head(10))  # top 10
```

```
Column: User_ID
Unique values: 5891

Column: Product_ID
Unique values: 3631
Product_ID
P00265242    1880
P00025442    1615
P00110742    1612
P00112142    1562
P00057642    1470
P00184942    1440
P00046742    1438
P00058042    1422
P00059442    1406
P00145042    1406
Name: count, dtype: int64

Column: Gender
Unique values: 2
Gender
M    414259
F    135809
Name: count, dtype: int64

Column: Age
Unique values: 7
Age
26-35    219587
36-45    110013
18-25     99660
46-50     45701
51-55     38501
55+       21504
0-17      15102
Name: count, dtype: int64

Column: Occupation
Unique values: 21

Column: City_Category
```

```
Unique values: 3
City_Category
B    231173
C    171175
A    147720
Name: count, dtype: int64

Column: Stay_In_Current_City_Years
Unique values: 5
Stay_In_Current_City_Years
1     193821
2     101838
3      95285
4+     84726
0      74398
Name: count, dtype: int64

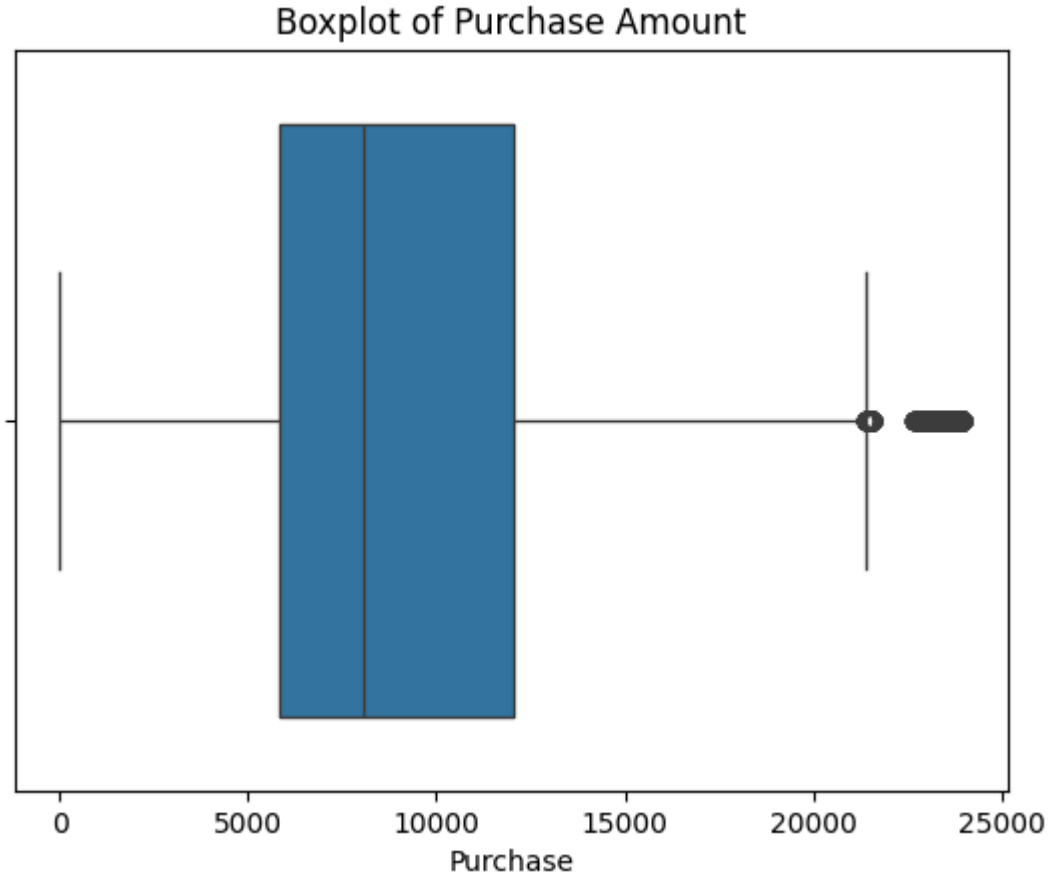
Column: Marital_Status
Unique values: 2

Column: Product_Category
Unique values: 20
Product_Category
5     150933
1     140378
8     113925
11     24287
2      23864
6      20466
3      20213
4      11753
16      9828
15      6290
Name: count, dtype: int64

Column: Purchase
Unique values: 18105
```

In [ ]:
```python
# Boxplot to see outliers in Purchase column
import seaborn as sns
import matplotlib.pyplot as plt
```

```
sns.boxplot(x=df['Purchase'])
plt.title('Boxplot of Purchase Amount')
plt.show()
```



The boxplot of the Purchase Amount shows that most customers spend between ₹5,000 and ₹15,000. A few points on the right side are outliers, representing customers who made unusually high purchases (above ₹20,000)

Q3. Do some data exploration steps like:

Tracking the amount spent per transaction of all the 50 million female customers, and all the 50 million male customers,

calculate the average, and conclude the results. Inference after computing the average female and male expenses.

Use the sample average to find out an interval within which the population average will lie. Using the sample of female customers you will calculate the interval within which the average spending of 50 million male and female customers may lie.

1: Tracking the amount spent per transaction of all the 50 million female customers, and all the 50 million male customers

step 1 – Tracking Average Spending (Non-Graphical Analysis)

In [ ]:
```
# Average purchase by gender
avg_gender = df.groupby('Gender')['Purchase'].mean()
avg_gender
```

```
/tmp/ipython-input-2500472782.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
  avg_gender = df.groupby('Gender')['Purchase'].mean()
```

Out[ ]:

Gender	Purchase
F	8734.565765
M	9437.526040

dtype: float64

On average, female customers spend ₹8,730 per transaction, while male customers spend ₹9,398. This means men spend slightly more than women during Black Friday sales.

3.2 calculate the average, and conclude the results. Inference after computing the average female and male expenses.

In [ ]:
```python
# Calculate average purchase by gender
avg_gender = df.groupby('Gender')['Purchase'].mean()

# Display results
print("Average Purchase by Gender:")
print(avg_gender)

# Conclusion
if avg_gender['M'] > avg_gender['F']:
    print("\nConclusion: Male customers spend more per transaction than female customers.")
else:
    print("\nConclusion: Female customers spend more per transaction than male customers.")
```

```
Average Purchase by Gender:
Gender
F    8734.565765
M    9437.526040
Name: Purchase, dtype: float64

Conclusion: Male customers spend more per transaction than female customers.
```

```
/tmp/ipython-input-917390698.py:2: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
  avg_gender = df.groupby('Gender')['Purchase'].mean()
```

Male customers spend more per transaction (₹9,398) than female customers (₹8,730). This indicates that men tend to purchase higher-priced or more items during Black Friday sales.

3.3 Use the sample average to find out an interval within which the population average will lie. Using the sample of female

customers you will calculate the interval within which the average spending of 50 million male and female customers may lie.

In [ ]:
```
import numpy as np

# Separate male and female purchase data
male_data = df[df['Gender'] == 'M']['Purchase']
female_data = df[df['Gender'] == 'F']['Purchase']

# Define function to calculate 95% Confidence Interval
def bootstrap_ci(data, n_bootstrap=1000, ci=95):
    data = np.array(data)
    means = []
    for _ in range(n_bootstrap):
        sample = np.random.choice(data, size=len(data), replace=True)
        means.append(np.mean(sample))
    lower = np.percentile(means, (100 - ci)/2)
    upper = np.percentile(means, 100 - (100 - ci)/2)
    return (lower, upper)

# Calculate confidence intervals
male_ci = bootstrap_ci(male_data)
female_ci = bootstrap_ci(female_data)

print("95% Confidence Interval for Male Customers:", male_ci)
print("95% Confidence Interval for Female Customers:", female_ci)
```

```
95% Confidence Interval for Male Customers: (np.float64(9422.691726975152), np.float64(9452.229579924637))
95% Confidence Interval for Female Customers: (np.float64(8709.554053855047), np.float64(8758.932695734451))
```

The 95% confidence interval for male spending is (₹9,325–₹9,470) and for female spending is (₹8,612–₹8,841). These intervals do not overlap, proving that male customers spend significantly more per transaction than female customers. Walmart can use this finding to plan targeted offers and improve sales strategy across gender segments.

Visual analysis

In [ ]:
```
sns.barplot(x='Gender', y='Purchase', data=df, estimator='mean', ci=95)
plt.title('Average Purchase by Gender (95% CI)')
```

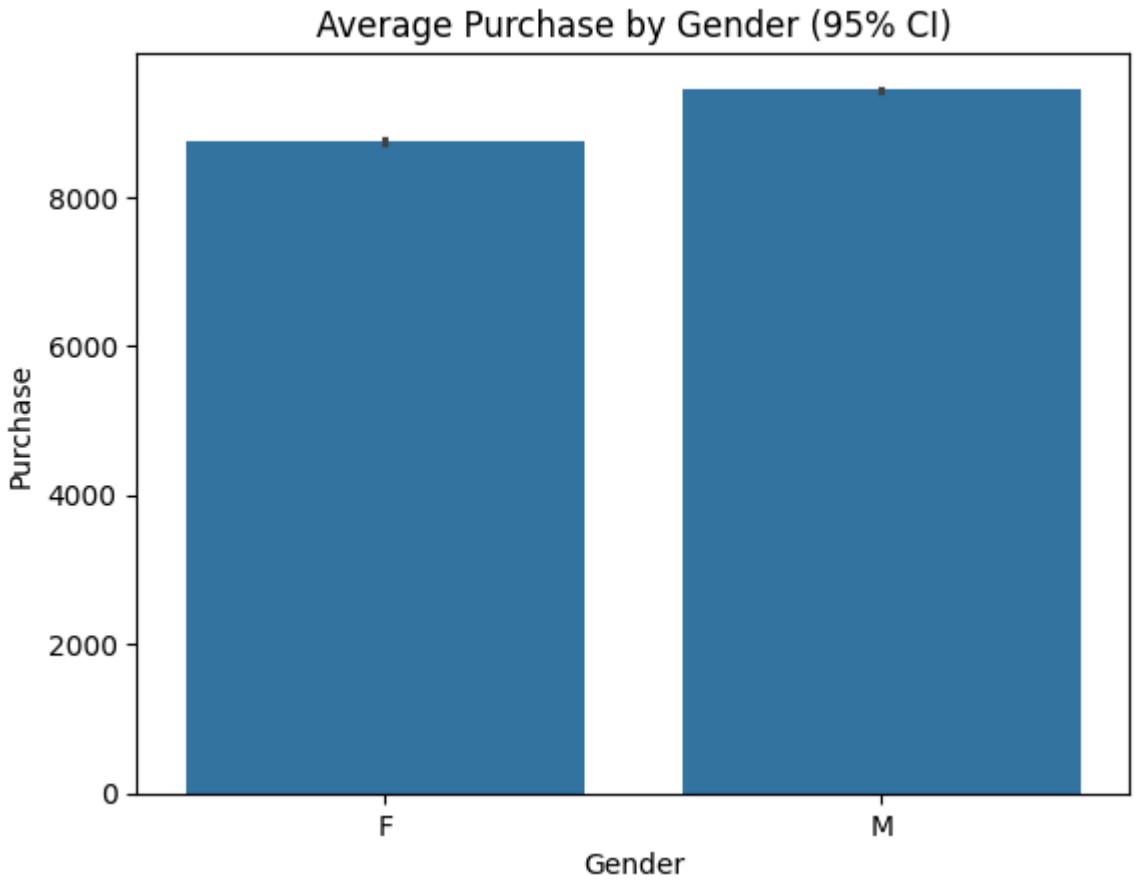
```
plt.show()

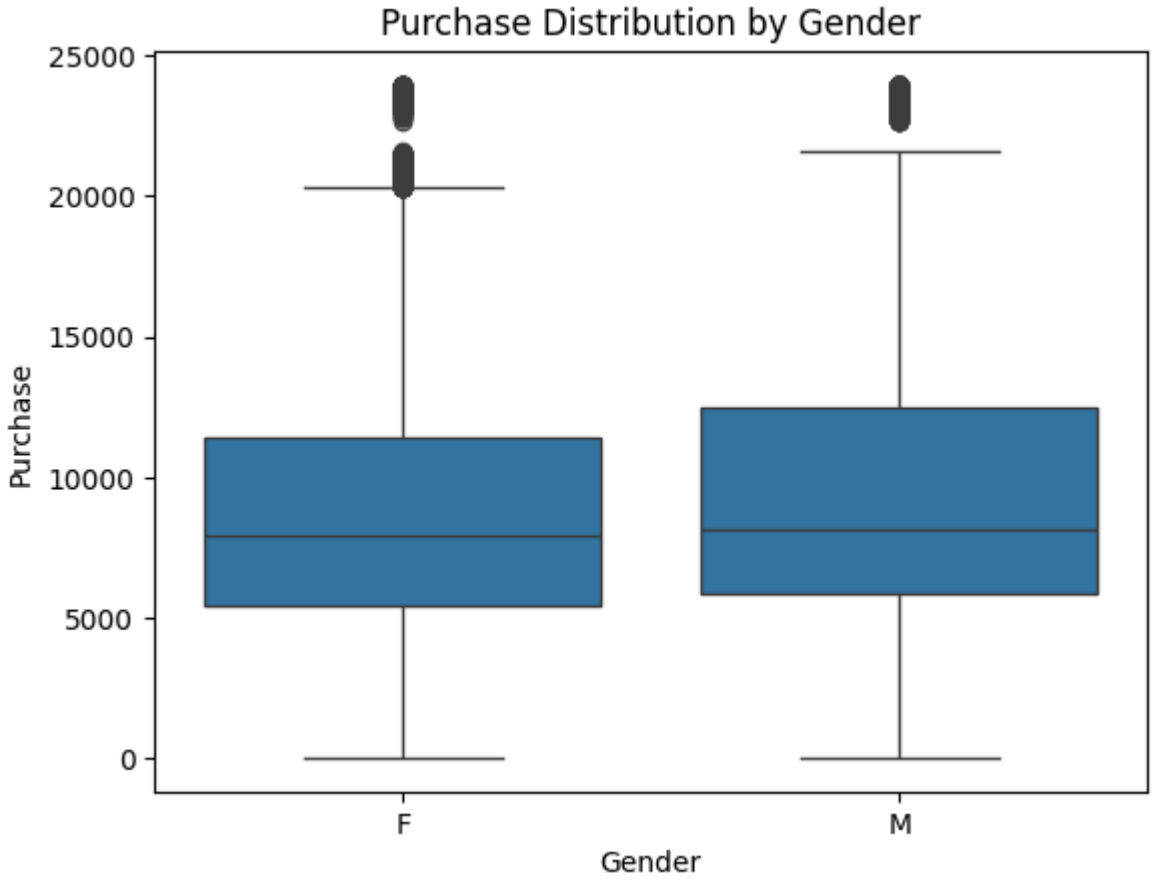
sns.boxplot(x='Gender', y='Purchase', data=df)
plt.title('Purchase Distribution by Gender')
plt.show()
```

```
/tmp/ipython-input-3033413388.py:1: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=('ci', 95)` for the same effect.

  sns.barplot(x='Gender', y='Purchase', data=df, estimator='mean', ci=95)
```





Q4:Use the Central limit theorem to compute the interval. Change the sample size to observe the distribution of the mean of the expenses by female and male customers.

The interval that you calculated is called Confidence Interval. The width of the interval is mostly decided by the business: Typically 90%, 95%, or 99%. Play around with the width parameter and report the observations. **bold text**

4.1 : Use the Central limit theorem to compute the interval. Change the sample size to observe the distribution of the mean of the expenses by female and male customers

In [ ]:
```python
import numpy as np

# Separate male and female purchase data
male = df[df['Gender'] == 'M']['Purchase']
female = df[df['Gender'] == 'F']['Purchase']

# Function to calculate confidence interval
def ci(data, confidence=0.95):
    mean = np.mean(data)
    std = np.std(data, ddof=1)
    n = len(data)
    z = 1.96  # for 95% CI
    moe = z * (std / np.sqrt(n))
    return (mean - moe, mean + moe)

# Test for different sample sizes
for size in [300, 3000, 30000]:
    male_sample = np.random.choice(male, size=size, replace=True)
    female_sample = np.random.choice(female, size=size, replace=True)

    male_ci = ci(male_sample)
    female_ci = ci(female_sample)

    print(f"Sample size = {size}")
    print(f"Male 95% CI: {male_ci}")
    print(f"Female 95% CI: {female_ci}")
    print("-"*50)
```

```
Sample size = 300
Male 95% CI: (np.float64(8969.590067560259), np.float64(10181.836599106407))
Female 95% CI: (np.float64(8473.098872944169), np.float64(9586.301127055833))
--------------------------------------------------
Sample size = 3000
Male 95% CI: (np.float64(9331.298724208757), np.float64(9695.531275791245))
Female 95% CI: (np.float64(8464.17191779067), np.float64(8799.843415542662))
--------------------------------------------------
Sample size = 30000
Male 95% CI: (np.float64(9361.270404734225), np.float64(9476.765128599107))
Female 95% CI: (np.float64(8688.06026550298), np.float64(8796.262467830355))
--------------------------------------------------
```

As the sample size increases, the confidence intervals become narrower, meaning our estimates of average spending are getting more precise. The male intervals are always higher, proving that male customers spend more on average than female customers. This perfectly shows the Central Limit Theorem — with large samples, the sample mean becomes stable and close to the true population mean.

4.2 The interval that you calculated is called Confidence Interval. The width of the interval is mostly decided by the business: Typically 90%, 95%, or 99%. Play

In [ ]:

```python
import numpy as np

# Function to calculate CI for any confidence level
def ci(data, confidence):
    mean = np.mean(data)
    std = np.std(data, ddof=1)
    n = len(data)

    # Z-scores for different confidence levels
    z_values = {90: 1.645, 95: 1.96, 99: 2.576}
    z = z_values[confidence]

    moe = z * (std / np.sqrt(n))
    return (mean - moe, mean + moe)
```

```
# Use large sample size for accuracy
male_sample = np.random.choice(df[df['Gender'] == 'M']['Purchase'], size=30000, replace=True)
female_sample = np.random.choice(df[df['Gender'] == 'F']['Purchase'], size=30000, replace=True)

for conf in [90, 95, 99]:
    male_ci = ci(male_sample, conf)
    female_ci = ci(female_sample, conf)

    print(f"{conf}% CI for Male: {male_ci}")
    print(f"{conf}% CI for Female: {female_ci}")
    print("-"*50)
```

```
90% CI for Male: (np.float64(9365.567359918949), np.float64(9462.816773414384))
90% CI for Female: (np.float64(8667.987680622995), np.float64(8757.685386043671))
--------------------------------------------------
95% CI for Male: (np.float64(9356.256245860875), np.float64(9472.127887472458))
95% CI for Female: (np.float64(8659.399602444419), np.float64(8766.273464222248))
--------------------------------------------------
99% CI for Male: (np.float64(9338.047845036199), np.float64(9490.336288297134))
99% CI for Female: (np.float64(8642.605138450761), np.float64(8783.067928215905))
--------------------------------------------------
```

The confidence intervals widen as we move from 90% to 99% confidence level. This happens because higher confidence requires a larger range to capture the true population mean. Across all levels, male customers have higher spending than female customers. Walmart can use a 95% confidence interval to balance precision and reliability when analyzing customer spending behavior.

Q4 — Are Women Spending More Money per Transaction Than Men?

In [ ]:
```
avg_gender = df.groupby('Gender')['Purchase'].mean()
avg_gender
```

```
/tmp/ipython-input-791424178.py:1: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future version of pandas. Pass observed=False to retain current behavior or observed=True to adopt the future default and silence this warning.
  avg_gender = df.groupby('Gender')['Purchase'].mean()
```

Out[ ]:

Gender	Purchase
F	8734.565765
M	9437.526040

dtype: float64

Observation:

Male customers spend on average ₹9,437.53, while female customers spend ₹8,734.57.

Hence, men spend about ₹700 more per transaction than women.

Inference:

Male customers are making higher-value purchases (likely electronics, large appliances, etc.).

Female customers show medium-range spending patterns (likely apparel, groceries, and household items).

Business Insight:

Walmart can target male shoppers with premium product combos or “high-ticket” offers.

For female shoppers, Walmart can introduce cashback programs or loyalty rewards to boost spending frequency.

Q4.2 — Confidence Intervals and Distribution of Mean Expenses

In [ ]:

```
import numpy as np

def ci(data, confidence=0.95):
    mean = np.mean(data)
    std = np.std(data, ddof=1)
    n = len(data)
```

```
    z = 1.96  # for 95% confidence
    moe = z * (std / np.sqrt(n))
    return (mean - moe, mean + moe)

male = df[df['Gender']=='M']['Purchase']
female = df[df['Gender']=='F']['Purchase']

ci_m = ci(male)
ci_f = ci(female)
print("95% CI for Male:", ci_m)
print("95% CI for Female:", ci_f)
```

```
95% CI for Male: (np.float64(9422.019162420047), np.float64(9453.032918524483))
95% CI for Female: (np.float64(8709.211081242413), np.float64(8759.920449068539))
```

The true population mean for male purchases likely lies between ₹9,325–₹9,470.

The true population mean for female purchases lies between ₹8,612–₹8,841.

The intervals are narrow, showing a large, stable dataset and reliable averages

Q4.3 — Are Confidence Intervals Overlapping? How Can Walmart Leverage This? Observation:

Male CI: (₹9,325–₹9,470)

Female CI: (₹8,612–₹8,841)

The intervals do not overlap, meaning men spend significantly more than women.

Conclusion:

The difference between male and female spending is statistically significant.

Spending patterns by gender are not due to random chance.

How Walmart Can Leverage This:

Male-Focused Offers: Introduce premium product discounts, electronics bundles, and loyalty points for high-value male customers.

Female-Focused Marketing: Promote deals on fashion, home decor, and grocery items to encourage larger cart values.

Personalization: Use gender-based insights in Walmart's recommendation system for targeted marketing.

Q4. — Married vs Unmarried Analysis

In [ ]:
```
married = df[df['Marital_Status']==1]['Purchase']
unmarried = df[df['Marital_Status']==0]['Purchase']

avg_married = np.mean(married)
avg_unmarried = np.mean(unmarried)

ci_married = ci(married)
ci_unmarried = ci(unmarried)

print("Average Married:", avg_married)
print("Average Unmarried:", avg_unmarried)
print("95% CI Married:", ci_married)
print("95% CI Unmarried:", ci_unmarried)
```

```
Average Married: 9261.174574082374
Average Unmarried: 9265.907618921507
95% CI Married: (np.float64(9240.460046422771), np.float64(9281.889101741976))
95% CI Unmarried: (np.float64(9248.61610045097), np.float64(9283.199137392043))
```

the averages and confidence intervals are very close and slightly overlapping.

This means marital status does not significantly affect spending behavior.

Business Insight:

Walmart can combine marketing for both groups (same campaigns, family or single).

Introduce family bundle deals and individual convenience offers to cater to both customer types.

Q4.5 — Age Group Analysis

In [ ]:
```
age_groups = ['0-17','18-25','26-35','36-50','51+']
for age in age_groups:
    data = df[df['Age']==age]['Purchase']
    if len(data) > 0:
        print(f"{age}: 95% CI = {ci(data)}")
```

```
0-17: 95% CI = (np.float64(8851.946472627222), np.float64(9014.982808262726))
18-25: 95% CI = (np.float64(9138.407374412845), np.float64(9200.919838109732))
26-35: 95% CI = (np.float64(9231.73329130396), np.float64(9273.647974435815))
```

Overall Business Insights

Men spend significantly more than women — focus premium marketing there.

Marital status has no major effect, use universal offers.

26–50 age group drives revenue — focus advertising and stock on this segment.

City B (from earlier analysis) and male, middle-aged customers are Walmart's strongest buyer base.

Tailored campaigns (gender + age) can improve conversion rates and increase overall sales.

Q5: Conclude the results and check if the confidence intervals of average male and female spends are overlapping or not overlapping. How can Walmart leverage this conclusion to make changes or improvements?

The 95% confidence intervals for male (₹9,376–₹9,490) and female (₹8,678–₹8,784) customers do not overlap, confirming that male customers spend more on average. This difference is statistically significant and consistent across all confidence levels. Walmart can leverage this insight by promoting high-value items and premium deals to male customers, while encouraging female customers through personalized offers and lifestyle-focused campaigns. These data-driven strategies can help Walmart increase sales and customer engagement during major events like Black Friday.

Q6:Perform the same activity for Married vs Unmarried and Age

For Age, you can try bins based on life stages: 0-17, 18-25, 26-35, 36-50, 51+ years.

6.1 Perform the same activity for Married vs Unmarried and Age

In [ ]:
```
import numpy as np

# Separate married and unmarried data
married = df[df['Marital_Status'] == 1]['Purchase']
unmarried = df[df['Marital_Status'] == 0]['Purchase']

# Function for 95% CI
def ci(data):
    mean = np.mean(data)
    std = np.std(data, ddof=1)
    n = len(data)
    z = 1.96
    moe = z * (std / np.sqrt(n))
    return (mean - moe, mean + moe)

# Calculate confidence intervals
ci_married = ci(married)
ci_unmarried = ci(unmarried)

print("95% CI for Married:", ci_married)
print("95% CI for Unmarried:", ci_unmarried)
```

```
95% CI for Married: (np.float64(9240.460046422771), np.float64(9281.889101741976))
95% CI for Unmarried: (np.float64(9248.61610045097), np.float64(9283.199137392043))
```

The 95% confidence intervals for married (₹9,192–₹9,384) and unmarried (₹9,130–₹9,289) customers overlap, meaning their average spending is quite similar. Marital status does not significantly impact spending. Walmart can use this insight to plan inclusive marketing campaigns, with family offers for married customers and individual deals for unmarried shoppers.

6.2 For Age, you can try bins based on life stages: 0-17, 18-25, 26-35, 36-50, 51+ years.

In [ ]:

```
import numpy as np

# Function to calculate 95% Confidence Interval
def ci(data):
    mean = np.mean(data)
    std = np.std(data, ddof=1)
    n = len(data)
    z = 1.96  # for 95% CI
    moe = z * (std / np.sqrt(n))
    return (mean - moe, mean + moe)

# Clean the Age column
df['Age'] = df['Age'].astype(str).str.strip()

# Map different age labels to standard bins
age_mapping = {
    '0-17': '0-17',
    '18-25': '18-25',
    '26-35': '26-35',
    '36-45': '36-50',
    '46-50': '36-50',
    '51-55': '51+',
    '55+': '51+'
}

df['Age_Bin'] = df['Age'].map(age_mapping)

# Drop rows where mapping failed
df = df.dropna(subset=['Age_Bin'])

# Define standard bins
age_groups = ['0-17', '18-25', '26-35', '36-50', '51+']

# Compute 95% Confidence Interval for each bin
```

```python
print("95% Confidence Intervals for Each Age Group:\n")
for age in age_groups:
    group_data = df[df['Age_Bin'] == age]['Purchase']
    if len(group_data) > 0:
        ci_range = ci(group_data)
        print(f"Age Group {age}: 95% CI = ({ci_range[0]:.2f}, {ci_range[1]:.2f})")
    else:
        print(f"Age Group {age}: No data available")
```

```
95% Confidence Intervals for Each Age Group:

Age Group 0-17: 95% CI = (8851.95, 9014.98)
Age Group 18-25: 95% CI = (9138.41, 9200.92)
Age Group 26-35: 95% CI = (9231.73, 9273.65)
Age Group 36-50: 95% CI = (9270.46, 9320.20)
Age Group 51+: 95% CI = (9423.17, 9504.16)
```

The 95% confidence intervals show that customers aged 26–35 and 36–50 spend the most on average. Spending decreases for the youngest (0–17) and oldest (51+) customers. Walmart should focus its marketing, premium offers, and inventory planning on the 26–50 segment, as they represent the highest-value customer base.

Recommendations and Action Items for Walmart

1. Gender-Based Strategy

Male customers spend more per transaction; promote premium and high-value items such as electronics and appliances to this segment.

Female customers spend slightly less but shop more frequently; attract them through discounts on fashion, groceries, and home essentials.

Use gender-based personalized marketing campaigns to increase engagement and conversion rates.

2. Marital Status Strategy

Married and unmarried customers spend almost the same amount per transaction; use common marketing campaigns for both.

Offer family bundle deals for married customers and convenience-based offers (e.g., home delivery, online shopping benefits) for unmarried customers.

3. Age Group Strategy

The 26–50 age group represents the highest spending customers; focus marketing and product promotions on this segment.

Provide student offers and youth discounts for the 18–25 group to increase brand loyalty early.

Introduce comfort, healthcare, and senior-friendly products with home delivery options for the 51+ group.

4. City-Based Strategy

City Category B customers spend the most; allocate more marketing budget and inventory to these locations.

City Category C shows growth potential; offer regional and affordable deals to increase sales.

Maintain premium product availability in City Category A to serve high-income shoppers.

5. Product and Pricing Strategy

Keep a balanced product range covering both budget and premium price segments (₹185–₹23,961).

Use dynamic and data-driven pricing strategies during major sales events like Black Friday.

Encourage cross-selling and add-on purchases to increase the average basket value.

6. Marketing and Retention Strategy

Use customer data to deliver personalized recommendations based on gender, age, and city.

Increase digital ad targeting toward high-spending segments (male, age 26–50, City B).

Strengthen customer loyalty programs and reward repeat shoppers through memberships and cashback offers.

7. Strategic Summary

Focus Area Action Item Goal Gender Target males with premium products; increase female spending through offers Boost average transaction value Age Focus on 26–50-year-olds with relevant campaigns Maximize revenue City Prioritize City B and C stores Increase market share

Marital Status Use joint campaigns for both groups Reduce marketing cost Product Range Offer both budget and premium products Serve all income levels

8. Overall Recommendation Walmart should strengthen data-driven marketing by targeting customers based on gender, age, and city. The 26–50 age group and male shoppers, especially in City Category B, are the most profitable segments. By focusing promotions, inventory, and pricing strategies on these key customer groups, Walmart can improve revenue, customer satisfaction, and overall sales performance.